Data Set Description

DataFile1_G3.txt

- Includes the outputs used to make Figures 2, 5, 9, and 10, which are map figures of the geophysical characteristics and average hydrate/gas formation in the study area
- Columns: node number, longitude, latitude, sedimentation rate, TOC, heat flux, average hydrate formed, average gas formed, number of simulations where gas saturation reached 1% and 2%
- Can be accessed using Numpy's loadtxt function
- 6048 rows of tab delimited data

DataFile2_G3.txt

- Includes the outputs used to make Figures 7 and 8, which are map figures of the correlation coefficients
- Columns: node number, longitude, latitude, correlation coefficients for hydrate formation with sedimentation rate, TOC, heat flux, and methanogenesis rate and the maximum of these four correlation coefficients at each site
- Can be accessed using Numpy's loadtxt function
- 6048 rows of tab delimited data

FigureData.h5

- HDF5 formatted file with profile data used to make Figures 3, 4, 6, and 11, which are a combination of sediment profiles and hydrate mass formation map data
- Organized by Groups based on the Figures
    - Figure_03 has profiles of Hydrate and Gas Saturation, Temperature, Pressure, Porosity, and Mole Fraction for a single site (Profile 7675)
    - Figure_04 has profiles of Hydrate Saturation and Porosity for three sites (Profiles 5233, 6576, and 8099) that can be used to calculate the mass of hydrate formed in each simulation
    - Figure_06 has mass of Hydrate formed at each location in the study area for 12 different simulation sets (9072 total locations, study area is locations 3025-9072)
    - Figure_11 has the Gas and Hydrate Saturation Profiles for the ODP Site 997 simulations run to compare to results from the Bhatanagar et al., 2007 publication. The simulations with the exact match to the Bhatnagar parameters are stored as Exact_Hydrate and Exact_Gas
- Can be accessed with H5PY module

Example Code

```
import numpy as np

import h5py as hf

# Load all data from DataFile1_G3.txt into D1

D1 = np.loadtxt('DataFile1_G3.txt',skiprows=1)

# Load all data from DataFile2_G3.txt into D2

D2 = np.loadtxt('DataFile2_G3.txt',skiprows=1)

# Load all data from FigureData.h5

F = hf.File('FigureData.h5')

# Access data in the Figure_06 iterations

M = F['Figure_06']['Hydrate_Mass_03'][:]

# Reshape the data for plotting purposes

M2 = np.reshape(M,(108,84));

# Access data in the Figure_11 Bhatnagar plots

H = F['Figure_11']['Bhatnagar']['Hydrate_16'][:]
```